



Independent experimental measurements of diffusion, sorption, and permeability support the solution-diffusion model of membrane transport

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ABSTRACT

The solution-diffusion (SD) model remains the cornerstone framework for interpreting mass transport in nonporous polymeric membranes, yet its applicability has recently been scrutinized. In this study, we highlight that independent measurement of the SD parameters can be reliably utilized to make reasonable flux predictions across many different types of transport modalities; in this article, we highlight three: hydraulic permeation and organic solvent reverse osmosis (OSRO), pervaporation, and vapor permeation. First, independent measurements of sorption (equilibrium uptake under varying penetrant molecule fugacities) and diffusion (from pulsed field gradient nuclear magnetic resonance) were used to parameterize the SD model. Afterward, the SD predictions of membrane transport were compared with independent experimental permeation data. Our findings demonstrate that, when independently measured sorption and diffusion coefficients are used to calculate permeation rates, the resulting values align closely with those obtained through direct permeation experiments across all three processes for different organic molecules and water. Our analysis underscores that the SD model provides a physically consistent and quantitatively accurate description of transport phenomena in dense membranes. This work reinforces the broad utility of the SD model and provides a framework for its validation in current and emerging membrane technologies.

1. Introduction

Selective separation of small molecules via the usage of nonporous dense membranes in application areas such as gas separations, pervaporation, seawater desalination, solvent recovery, crude oil fractionation, valuable solute and biochemical concentration, and others has been widely and accurately predicted by the solution-diffusion (SD) model for decades [1–8]. The solution-diffusion model (which itself is a special case of the more general sorption-diffusion model that is capable of also describing transport in microporous membranes) was first proposed by Thomas Graham in 1865 [9] and mathematically formulated by von Wroblewski in 1879 [10]. Since then, the SD model has gained acceptance due to its simplicity and accuracy in first predicting gas permeation processes and then later liquid phase solute/solvent permeation and separation. Complexities such as polymer swelling, non-ideal driving forces [11], non-constant diffusivities, non-linear isotherms, and others, which are not treated by the simplest formulation of the SD model, can also be handled in more rigorous forms of the SD model [11–14]. Importantly, correctly formulated SD models have

been shown to accurately predict hydraulic permeances and vapor permeations through defect-free dense membranes, a point that we will underscore in this article.

Recent studies have suggested that the solution-diffusion model does not apply to pressure-induced permeation of organic solvents in polymeric membranes [15–17]. For instance, Fan et al., 2024 and He et al., 2024 put forth a solution-friction model and show evidence that solvent transport through polymer membranes occurs through a viscous flow mechanism [15,16]. However, the conclusions of these articles rely heavily on molecular simulations of polymeric systems, and polymeric systems are known to have long relaxation times. These long relaxation times, relative to the timescales available in computational molecular models, suggest that the validity of the solution-friction mechanism relies heavily on the initial structure of the polymer as well as the size of the simulation system, which is typically on the order of tens of nanometers [18,19]. Moreover, the experimental evidence that is provided (such as the lack of curvature in flux vs pressure data) does not convincingly reject the solution-diffusion transport model, nor does it provide strong evidence of viscous flow (pore-flow) transport [4,20].

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Readers are referred to Refs. [11,20,21] for detailed mechanistic derivations of SD and how they give rise to phenomenological outcomes such as curved (or not curved) flux vs pressure data for hydraulic permeation experiments. In this article, we will reinforce experimental evidence of the underlying physics that govern solution-diffusion transport.

Indeed, one key challenge with the solution-friction theory is the difficulty in measuring the frictional coefficients needed to parameterize the solution-friction model. There are no known techniques to measure these coefficients outside of permeation experiments themselves, and thus, using permeation techniques to support the solution-friction model is a somewhat circular argument (i.e., rate-based experimental techniques are utilized to extract rate parameters).

Importantly, the parameters in the solution-diffusion model can each be measured independently using non-permeation techniques, and have been reliably used in predicting unary solvent permeation and multi-component mixture separation (including the accurate prediction of 400 components co-permeating through a membrane simultaneously in one recent example [22]) [4,13,23–30]. Fig. 1 summarizes the transport of organic solvents in a polymeric membrane under the solution-diffusion mechanism, which is the focus of our manuscript. At its essence, the solution-diffusion permeation process involves a local thermodynamic equilibrium between the membrane interface and the adjacent fluid mixture. The non-equilibrium but steady state permeation process of molecules through the polymer structure involves random walk diffusive jumps via transient gaps that form between the polymer chains. Over sufficiently long time scales, these jumps will follow normal diffusion time scaling [31,32]. These gaps form as a result of thermal fluctuations of the polymer backbone, and the frequency at which a gap of size sufficient enough for the permeating molecule to take a diffusive jump is related to the size of that molecule. Importantly, the diffusive

jumps are not oriented (provided the membrane is an isotropic medium), and the permeating molecules move in a random walk but with a net flux due to the chemical potential difference placed across the membrane to induce permeation. We re-emphasize these solution-diffusion fundamentals, as these two processes – thermodynamic partitioning from one phase to another, and random walk of a compound within a polymer – can be measured using equilibrium, non-permeation-based techniques.

The main point of this article is that both the sorption isotherm and the diffusion coefficient can be measured using equilibrium techniques, which reduces the significant uncertainties often associated with extracting kinetic rate constants from permeation experimentation. Afterward, these two equilibrium-measured parameters can be combined using the appropriate form of the solution-diffusion expression. Here, we utilize the Maxwell-Stefan (MS) solution-diffusion formulation as originally formulated by Krishna et al. [33,34] to predict the steady-state hydraulic flux through a dense membrane layer (Riberio et al. also developed a Maxwell-Stefan form of the solution-diffusion equations [35], but we find the Krishna form easier to implement). This MS formulation states that the penetrant flux through a dense selective layer is governed by the penetrant-polymer diffusion coefficients, sorption interaction within the membrane, and the membrane-phase volume fraction gradient (simplified from chemical potential gradient) across the membrane. In this paper, we highlight that with knowledge of the constitutive parameters involved in the solution-diffusion model, it is possible to verify the accuracy of the SD model in predicting experimental liquid and vapor permeation rates through dense membranes.

2. Parameterization of the solution-diffusion model

According to the solution-diffusion model, organic solvent transport

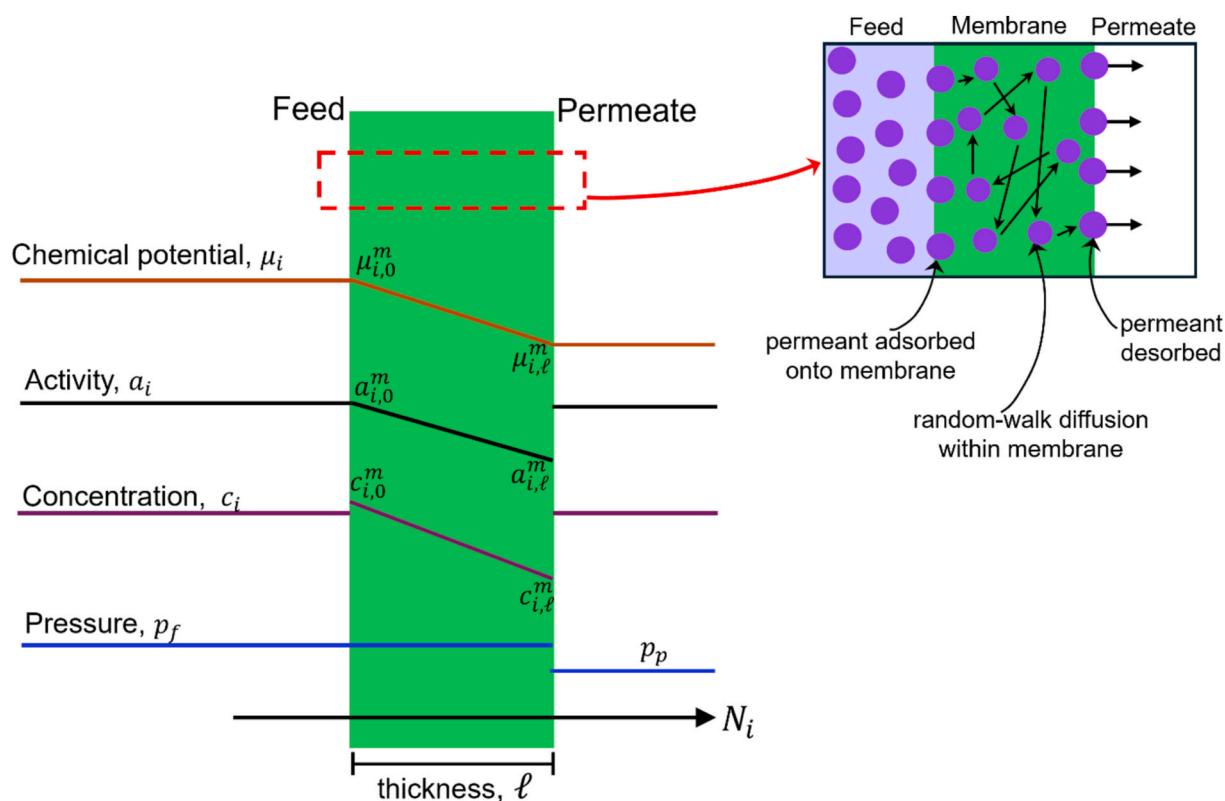


Fig. 1. Illustrative diagram of solvent's chemical potential, μ_i , activity, a_i , and concentration, c_i , through a nonporous membrane in the solution-diffusion (SD) mechanism. The feed and permeate-side solvent properties are denoted with subscripts, 0 and ℓ , respectively. Hydraulic solvent flux, N_i , through the dense membrane, is influenced by a pressure-induced concentration gradient driving force. By this, the permeant molecules are first sorbed onto the membrane feed side due to chemical affinity with membrane polymer, then undergo diffusional jumps across the film thickness via random thermal motion of the polymer chains (inset). This process occurs until the penetrant molecule reaches the permeate side before being desorbed into bulk permeate.

in dense polymeric membranes is based on the combination of a component's sorption coefficient, S_i (note, many authors confuse sorption or solubility coefficients with Hansen or Hildebrandt solubility coefficients) and its diffusion coefficient D_i . The simplest form of the model is often represented as in Eq. (1), where the solvent permeability, P_i , (flux normalized by driving force and membrane thickness) is the product of these two coefficients.

$$P_i = D_i \cdot S_i \quad (1)$$

A key parameter of the SD model is the solvent diffusivity in the polymer. The diffusion coefficients of penetrant molecules in membrane polymers can be obtained from a variety of experiments: sorption kinetic studies (e.g., time lag measurement or dynamic vapor sorption), pure component constant pressure permeation, proton nuclear magnetic resonance (NMR), pulse field gradient-NMR (PFG-NMR), or empirical equations such as the Stokes-Einstein equation [29,36–39]. Similar to extracting frictional coefficients from a pure component permeation experiment in the solution-friction model, the constant-pressure permeation experiment is a cyclical approach to estimating pure component diffusivity – i.e., the flux data is fit to a model to extract a rate parameter. Importantly, the membrane is under nonequilibrium conditions during these permeation experiments, which can introduce confounding experimental biases resulting from situations such as pressure-induced plasticization or compaction, deswelling on the membrane's downstream face, etc. However, in the PFG-NMR measurement, the membrane and the sorbed penetrant molecules are at equilibrium, and the technique allows for measurement of the displacement of the penetrant molecule within the polymer matrix in the absence of a permeation driving force. PFG-NMR techniques enable measurement of penetrant self-diffusivities inside of a closed system at chemical equilibrium via the Einstein relation,

$$\langle r^2(t) \rangle = 6D_{i, \text{self}} t \quad (2)$$

by measuring the mean squared displacement ($\langle r^2(t) \rangle$) of the guest such that it is smaller than the characteristic length of the membrane sample (i.e., the half-thickness) within the observation time, t . $D_{i, \text{self}}$ is the self-diffusion coefficient of the solvent species within the polymer phase.

In the PFG-NMR experiment, a pair of magnetic field gradient pulses is applied to encode and decode the positions of diffusing nuclei. A signal attenuation arises due to molecular displacements during the diffusion time t , causing a dephasing of the nuclear spins. This attenuation of the NMR signal Ψ , follows an exponential decay given as [36,37, 40]:

$$\Psi = \frac{S(g)}{S(g \approx 0)} = \exp(-D_{i, \text{self}} q^2 t) \quad (3)$$

Where Ψ is the PFG-NMR signal attenuation, S is the PFG-NMR signal intensity, $q = 2\gamma^* g \delta$ is the wavevector, where γ^* is the gyromagnetic ratio, g is the gradient pulse intensity (or gradient strength), and δ is the effective gradient pulse length. By varying g and measuring Ψ , $D_{i, \text{self}}$ can be extracted from the slope of the logarithmic attenuation curve.

Solvent sorption in polymers is generally measured via the gravimetric sorption method in dynamic vapor sorption instruments [4,41], quartz crystal microbalance equipment [42], or other custom-made sorption devices. Simplified soaking experiments are often utilized to estimate the mass and (sometimes) volume change of polymer films upon long-term immersion in liquid solvents. The solvent uptake in the polymer after long exposure times is treated as the equilibrium condition between the solvent and the polymer. The equilibrium mass fraction of the solvent in the membrane is directly estimated from gravimetric techniques, and the equilibrium volume fraction of the solvent in the membrane can be computed if corresponding dilatometry data are available or via reasonable assumptions of constant solvent molar volumes and isotropic dilation (if appropriate). Relating the solvent

concentration in the membrane to the relative pressure of the solvent (i.e., the partial pressure of the solvent in a carrier gas relative to the saturated vapor pressure of the solvent at the testing temperature), the solvent activity, or the solvent fugacity provides the equilibrium sorption isotherm for that solvent-polymer system. The isotherm encapsulates all physical and chemical interactions between the solvent and polymer over the activity range from ~ 0 (infinitely dilute solvent) to 1.0 (polymer submerged in solvent, often referred to as the “unit activity” condition). The isotherms can be fit with various sorption models such as the Henry isotherm, Langmuir, Flory-Huggins (or Flory-Rehner), or dual-mode sorption isotherms. Parameters in these isotherms have been linked to molecular-scale phenomena such as sorption in packing defects between glassy polymer chains [43–46]. Fig. 2 provides an overview of the solution-diffusion model parameters and the methods of measuring each parameter independently of rate-based techniques.

3. Formulation of the solution-diffusion model

Generally, the component molar flux, J_i , through a membrane is related to its chemical potential gradient driving force, $d\mu_i/dz$, as shown below.

$$J_i = -L_i \frac{d\mu_i}{dz} \quad (4)$$

where z is the coordinate of net molecular transport across the membrane thickness and L_i is the constant of proportionality. Considering concentration and pressure gradients as the main contributors to the chemical potential driving force, $d\mu_i$ can be represented as:

$$d\mu_i = RT d \ln(a_i) + \bar{V}_i dp \quad (5)$$

where a_i is the activity ($a_i = \gamma_i x_i$), p is pressure, and γ_i , x_i , and $\bar{V}_i$ are the activity coefficient, molar fraction, and partial molar volume of component i . Under the constant membrane pressure assumption of the SD model, Eq. (4) can be expressed as [47]:

$$J_i = -\frac{RTL_i}{x_i} \frac{dx_i}{dz} = -\frac{RTL_i}{c_i} \frac{dc_i}{dz} \quad (6)$$

where c_i is the molar concentration of component i . Eq. (6) is similar to Fick's law, and the component diffusivity can be defined as $D_i = \frac{RTL_i}{c_i}$, such that the flux equation becomes:

$$J_i = -D_i \frac{dc_i}{dz} = -D_{i, \text{Fick}} \frac{dc_i}{dz} \quad (7)$$

Here, we formulate the solution-diffusion model using the two-dimensional Maxwell-Stefan matrix form of Krishna [34] as follows:

$$(N^v)_{1,n} \int_0^L dz = - \int_{(\phi^o)_{1,n}}^{(\phi^m)_{1,n}} [B]^{-1} [I] d(\phi^m)_{1,n} \quad (8)$$

where $(N^v)_{1,n}$ is the vector of n penetrants' volumetric fluxes through a polymer membrane, ϕ^m is the $(n+1)$ dimensional vector of component volume fraction in the membrane phase including n penetrants and the polymer itself, and $[B]$ is the $(n \times n)$ dimensional matrix of the volume-based Maxwell-Stefan solvent-solvent ($D_{ij}^{v,m}$) and solvent-polymer diffusivities ($D_{i,n+1}^{v,m}$) and is represented as:

$$[B]_{ii} = \sum_{j=1}^n \frac{\phi_j^m}{D_{ij}^{v,m}} + \frac{\phi_{n+1}^m}{D_{i,n+1}^{v,m}}, [B]_{ij, i \neq j} = \frac{\phi_j^m}{D_{ij}^{v,m}} \quad (9)$$

$j \neq i$

The $(n \times n)$ dimensional sorption matrix, $[I]$, describes the solvent-solvent and solvent-polymer interactions and is defined as:

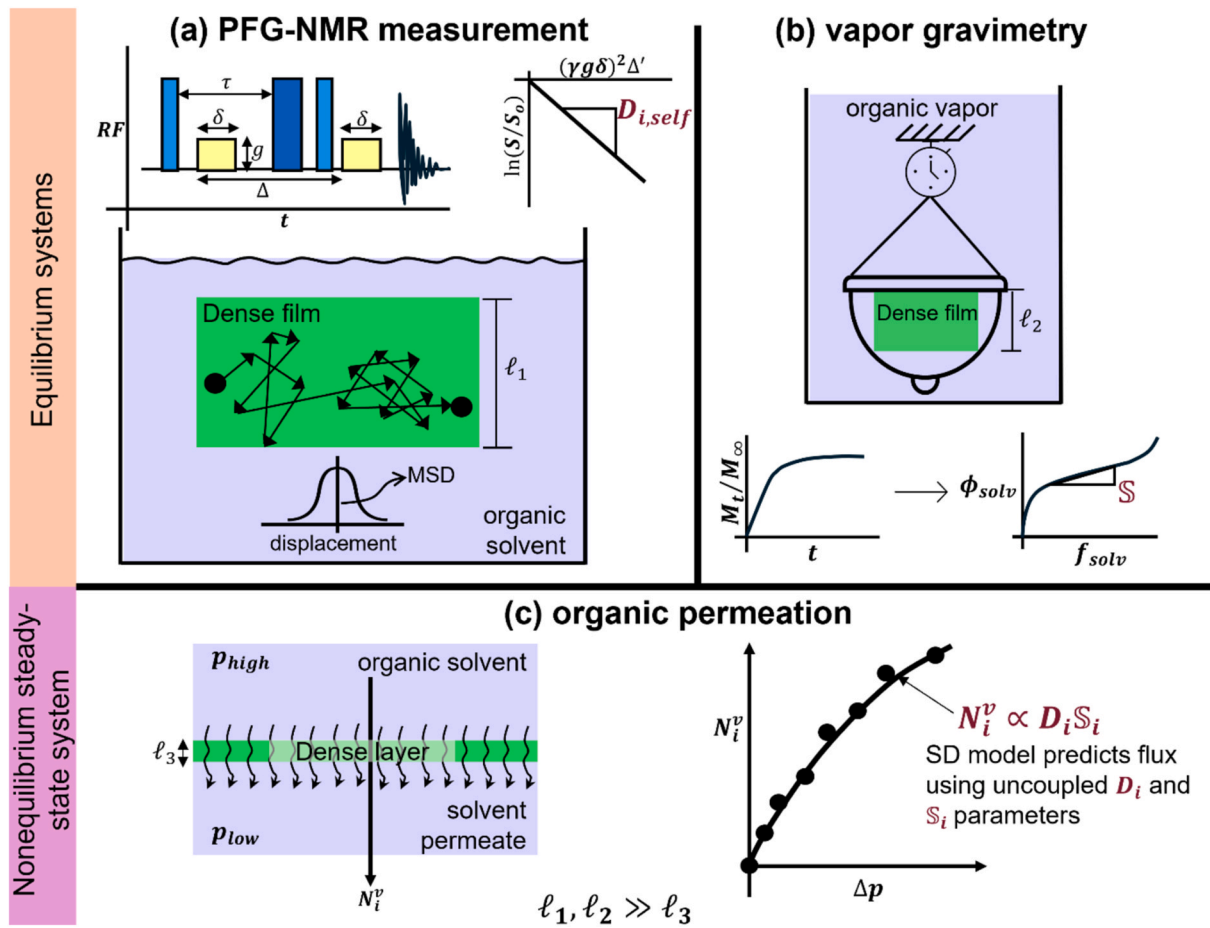


Fig. 2. Parameterization of the solution-diffusion model with experiments for each SD model parameter. (a) typical pulse field gradient-nuclear magnetic resonance (PFG NMR) experiment including signal attenuation plot for estimating the diffusion coefficient of organic solvent in membrane polymer, here MSD = mean squared displacement; (b) gravimetric vapor sorption experiment of polymer film to obtain sorption and sorption kinetics data at varying solvent fugacity (f_{solv}) or activity; the former data sets can be used to deduce the sorption coefficient, S ; (c) steady-state solvent permeation experiment in thin film composite organics solvent reverse osmosis membrane. In the SD mechanism, organic solvent flux through a dense membrane under the influence of a fugacity gradient driving force can be directly predicted from the solvent sorption coefficient and the solvent diffusivity in the membrane.

$$\Gamma_{ij} = \frac{d \ln a_i^m}{d \ln \phi_j^m} = \frac{\phi_i^m}{f_i^m} \frac{df_i^m}{d\phi_j^m} \quad (10)$$

where a_i^m is the sorbed solvent activity in the polymer phase and is defined as the ratio of the solvent fugacity in the membrane phase (f_i^m) to the fugacity of saturated vapor at feed temperature (f_i^0). In this article, we describe the solvent concentration in the membrane polymer by the Flory sorption isotherm with an extension by Rehner to account for the elastic contribution of swollen rubbery polymers. For single-component permeation, volumetric flux, N_i^v , Eq. (8) is reduced to:

$$N_i^v \int_0^{\phi_{1,\infty}^m} dz = - \int_{\phi_{1,0}^m}^{\phi_{1,\infty}^m} \frac{\bar{D}_{1,m}^{v,m}}{(1 - \phi_1^m)} \Gamma d\phi_1^m = \int_{\phi_{1,0}^m}^{\phi_{1,\infty}^m} \frac{\bar{D}_{1,m}^{v,m}}{(1 - \phi_1^m)} \frac{\phi_1^m}{f_1^m} \frac{df_1^m}{d\phi_1^m} d\phi_1^m \quad (11)$$

The Fickian diffusivity of the penetrant molecules, $D_{1m,\text{Fick}}$, in Eq. (7) is connected to the volume-based MS diffusivity, $\bar{D}_{1,m}^{v,m}$, as defined in Eq. (12) [34].

$$D_{1m,\text{Fick}} = \frac{1}{1 - \phi_1^m} \Gamma \bar{D}_{1,m}^{v,m} \quad (12)$$

Substituting Eq. (12) into Eq. (11) to introduce the Fickian diffusivity, which in the case of single solvent systems at unit activity, corresponds to the self-diffusivity of the penetrant molecule (measured by PFG-NMR experiment), gives:

$$N_i^v \int_0^{\phi_{1,\infty}^m} dz = -D_{1m,\text{Fick}} \int_{\phi_{1,0}^m}^{\phi_{1,\infty}^m} d\phi_1^m = -D_{1m,\text{self}} \int_{\phi_{1,0}^m}^{\phi_{1,\infty}^m} d\phi_1^m = -D_{1m,\text{PFG-NMR}} \int_{\phi_{1,0}^m}^{\phi_{1,\infty}^m} d\phi_1^m \quad (13)$$

where $D_{1m,\text{PFG-NMR}}$ is the penetrant-polymer diffusivity measured from the PFG-NMR experiment, and $\phi_{1,0}^m$ and $\phi_{1,\infty}^m$ are estimated from bulk and membrane phase activities via the sorption isotherms. Eq. (13) is used in this work to estimate the solvent flux through polymer membrane in the organic solvent reverse osmosis (OSRO) process.

While the usage of solvent phase fugacity is unambiguous, it is rarely utilized, and the activity of the solvent molecule in the upstream and downstream is typically used. A common approach to using the solvent-activity-based driving force is to utilize the high-pressure side of the membrane as the reference state for the activity calculations (i.e., $p_{\text{ref}} = p_{\text{high}} = p_o$). Based on this, the thermodynamic equilibrium in chemical potentials on the upstream membrane face results in $a_{i,o}^{\text{bulk}} = a_{i,o}^m$ (equal to 1 for pure component OSRO process) and yield the estimate of the solvent activity on the downstream membrane face as:

$$a_{i,\infty}^m = a_{i,\infty}^{\text{bulk}} \exp\left(-\bar{V}_i \frac{(p_o - p_r)}{RT}\right) \quad (14)$$

by assuming that a “pressure discontinuity” exists at the downstream

membrane face (where $\bar{V}_i$ is the penetrant partial molar volume and $p_{\downarrow}$ as the downstream pressure). This pressure discontinuity is counterintuitive at first and appears to be problematic from a mechanical point of view; however, it ultimately makes sense in the context of mechanical force balances. By this, the polymer matrix carries normal stress since there is no neighboring liquid column to transmit hydrostatic pressure through the film as in a porous medium. Thus, the interfacial force balances permit jumps in normal stress at phase boundaries provided the solid bears that load [48]. In addition, as demonstrated by Freger and Ramon 2024, in SD transport in thin-film composite membranes with thick active nonporous layers supported over narrow pores, the pressure across the membrane remains nearly uniform, and the solvent flux streamlines are evenly distributed [49]. Moreover, their calculations indicate that the pressure change is not a discontinuity but is instead a smooth function located near the downstream face of the membrane. This behavior emerges when the membrane film is uniformly loaded by feed pressure, and the influence of the support is confined to only a small near-pore region. The SD model thus remains physically consistent when mechanical equilibrium and realistic membrane support geometry are properly accounted for [49].

In the regime where the organic component's relative saturation is below unity, transport behavior in dense polymer membranes deviates from the commonly assumed concentration-independent diffusivity model. This deviation reflects a shift to concentration-dependent diffusion, which can be estimated using models such as the Mackie-Meares' equation [50], Long's equation [51], and Vrentas-Duda free volume theory [52], among others, in the absence of direct experimental D_i vs c_i or a_i data (the latter of which is preferred, but rarely available). This observation is particularly relevant in vapor permeation and pervaporation processes, where the bulk feed and/or permeate phases are often in the vapor state or at sub-saturation levels. The Mackie-Meares' equation is selected in this study due to its physical basis and low parameter needs (which enables its utilization via single-point diffusivity obtained from PFG-NMR measurement) [50].

According to the Mackie-Meares' model, the diffusivity of organic vapor in a polymer membrane, D_i^m , can be related to its volume fraction in the membrane phase as represented by Eq. (15) [50]:

$$D_i^m = D_i^{m0} \cdot \left(\frac{1 - \phi_p^m}{1 + \phi_p^m} \right)^2 = D_i^{m0} \cdot \left(\frac{\phi_i^m}{2 - \phi_i^m} \right)^2 \quad (15)$$

where D_i^{m0} is the penetrant self-diffusivity in its organic or aqueous phase as polymer volume fraction, ϕ_p^m approaches zero [52]. Substituting Eq. (15) into Eq. (13), the SD model in Eq. (16) can be easily used to estimate unary pervaporation and vapor permeation fluxes, viz.,

$$N_1^v \int_0^{\ell} dz = -D_1^{m0} \int_{\phi_{1,0}^m}^{\phi_{1,\ell}^m} \left(\frac{\phi_i^m}{2 - \phi_i^m} \right)^2 d\phi_i^m \quad (16)$$

For vapor permeation and pervaporation processes, the penetrant activities in the membrane phase can be estimated as in Eqs. (17) and (18) using the saturation vapor pressure of the solvent, p_1^{sat} as reference state.

$$a_{1,0}^m = a_{1,0}^{bulk} \exp \left(-\bar{V}_1 \frac{(p_o - p_1^{sat})}{RT} \right) \quad (17)$$

$$a_{1,\ell}^m = a_{1,\ell}^{bulk} \cdot \left(\frac{p_{1,\ell}}{p_1^{sat}} \right) \cdot \exp \left(-\bar{V}_1 \frac{(p_o - p_1^{sat})}{RT} \right) \quad (18)$$

where the bulk-phase activities are the ratio of partial pressures to the saturation vapor pressure. Under a deep vacuum application or high sweep gas rate on the downstream side of the membrane, an approximately zero downstream activity is achieved, and $\phi_{1,\ell}^m \ll \phi_{1,0}^m$. A flow-chart is provided in the supplementary information together with

worked examples showing the integration of these equations in flux prediction (Sections S5–6).

Table 1 presents the physical properties of solvents considered in this study. Solvent partial molar volume in the membrane phase, $\bar{V}_1$ is assumed to be the same as the liquid molar volume, $\hat{V}_1$, except for water in Nafion-117 (where $\bar{V}_1 \cong 17 \text{ cm}^3/\text{mol}$). At low hydration, water is strongly contracted by ion-dipole interactions with sulfonate groups of the membrane and by confinement/elastic constraints inside the ionic domains; only at very high hydration does it behave nearly like bulk water [53].

4. Solution-diffusion framework validated with independent experimental inputs

Independently sourced experimental data were utilized for model parameterization and validation to evaluate the predictive capability of the solution-diffusion model. Sorption isotherms were extracted from previously published literature [30,41,42,53–55] while self-diffusion coefficients were obtained from PFG-NMR measurements reported by Muzzalupo et al. [36]. The only choice for PFG-NMR diffusivity data for the organic solvents/PDMS system is the work of Muzzalupo et al. and are presented in Table S2. For the water/Nafion-117 system, the experimental data of Zawodzinski Jr. et al. [56] and Tsushima et al. [57] are used for comparison. The Arrhenius relation is used to adjust the water self-diffusivity of Zawodzinski Jr. et al. [56] to ambient temperature conditions as detailed in the Supplementary Information S3. We treat the unit-activity self-diffusion coefficients as the unit-activity Fickian diffusivities, and Eq. (13) was used for flux estimation. Permeation (flux) data used for model validation originated from different sets of unrelated studies on organic solvents such as n-heptane, n-hexane, toluene, acetone, cyclohexane, p-xylene, and water permeating through thin film composite membranes [15,58–60]. The membrane polymer considered in these studies is crosslinked polydimethylsiloxane (PDMS), except for water where proton-exchange membrane Nafion-117 is used due to its wide application in electrochemical systems. PDMS was selected for our analysis because there is a wealth of sorption, permeation, and PFG-NMR diffusometry data collected by polymer and membrane scientists over the last 30–40 years. The equilibrium data – sorption isotherms and PFG-NMR diffusion measurements of pure solvents – are taken from studies focusing on thick films (350–3500 μm) [30,41,42,53–55]. The Flory isotherm fittings of sorption data are presented in the Supplementary Information S1. To demonstrate the robustness of the solution-diffusion model, the predictions from the model with these independently measured equilibrium physical phenomena on thick films are compared against nonequilibrium but steady-state permeation data on thin film (1.5–3.0 μm active layer) composites and self-supporting films (180–200 μm). This approach was taken not only to minimize potential bias but also to test the robustness of the solution-diffusion framework when applied across heterogeneous

Table 1

Physical properties of organic solvents and water used in this study at ambient conditions ($\sim 24^\circ\text{C}$).

Solvents	Molecular weight, M_w (g/mol)	Density, ρ (kg/m ³)	Bulk liquid molar volume, $\hat{V}$ (cm ³ /mol)	Partial molar volume, $\bar{V}$ (cm ³ /mol)	Saturation vapor pressure, p^{sat} (kPa)
Toluene	92.14	867.0	106.27	106.27	3.60
Acetone	58.08	784.0	74.08	74.08	28.07
n-Heptane	100.20	679.5	147.46	147.46	5.80
n-Hexane	86.18	660.6	130.45	130.45	18.48
Cyclohexane	84.16	779.0	108.04	108.04	12.43
p-Xylene	106.16	856.7	123.92	123.92	10.52
Water	18.02	998.0	18.05	17.00	2.80

datasets. The transmembrane chemical potential or transmembrane fugacity – not the transmembrane pressure – was used to derive the SD fluxes (Eqs. (4)–(7)) as shown in Section 3, and a constant membrane pressure assumption was used as shown in Eq. (6). The dry membrane thicknesses provided in the literature were adjusted to account for the dimensional change upon swelling by using the relationship originally developed by Paul [11], viz. Eq. (19).

$$\ell_{\phi_1^m} = \ell_{dry} (1 - \phi_{1,o}^m)^{-1/3} \quad (19)$$

where ℓ_{dry} is the dry membrane thickness and $\phi_{1,o}^m$ is the solvent volume fraction on the upstream membrane face.

Fig. 3 shows the hydraulic permeation flux of various organic solvents in PDMS membrane as a function of transmembrane pressure using independent literature data for solvent sorption isotherm and diffusivity estimation for SD parameterization. We find that the solution-diffusion model can accurately predict the experimental flux of toluene (Fig. 3 (a)), acetone (b), and n-heptane (c), and *p*-xylene (f) without fitting flux data to extract transport parameters, as is required in the solution-friction model. For n-hexane and cyclohexane (Fig. 3(d)–(e)), the

experimental data lies closely outside the confidence interval of the SD prediction. Further details on the potential origin of this deviation from experimental data is provided in Section 5. We re-emphasize that the flux, diffusion, and sorption data are sourced from independent labs, each working on their own particular formulation of PDMS.

Despite the diversity of data sources, research groups, and membrane samples, the SD model accurately predicts experimental flux behavior, underscoring the reliability of the solution-diffusion model in describing organic solvent permeation across dense polymeric membranes. Whilst the recently proposed solution-friction model relies on frictional coefficient fitting from the same permeation experiment, the uniqueness of the SD model lies in its ability to implement independent solubility and diffusivity parameters to predict new and existing experimental permeation trends as well as multicomponent separation performance. For instance, Lee and coworkers integrated a machine learning predictor of pure component solvent uptakes and diffusivities into the solution-diffusion model and used the SD-based model for predicting the separation of a complex feed mixture containing up to 400 components through polymer membranes [22]. This was achieved with an excellent prediction accuracy (less than 8 % error) for crude oil fractionation.

Fig. 4 presents water permeation predicted by the solution–diffusion

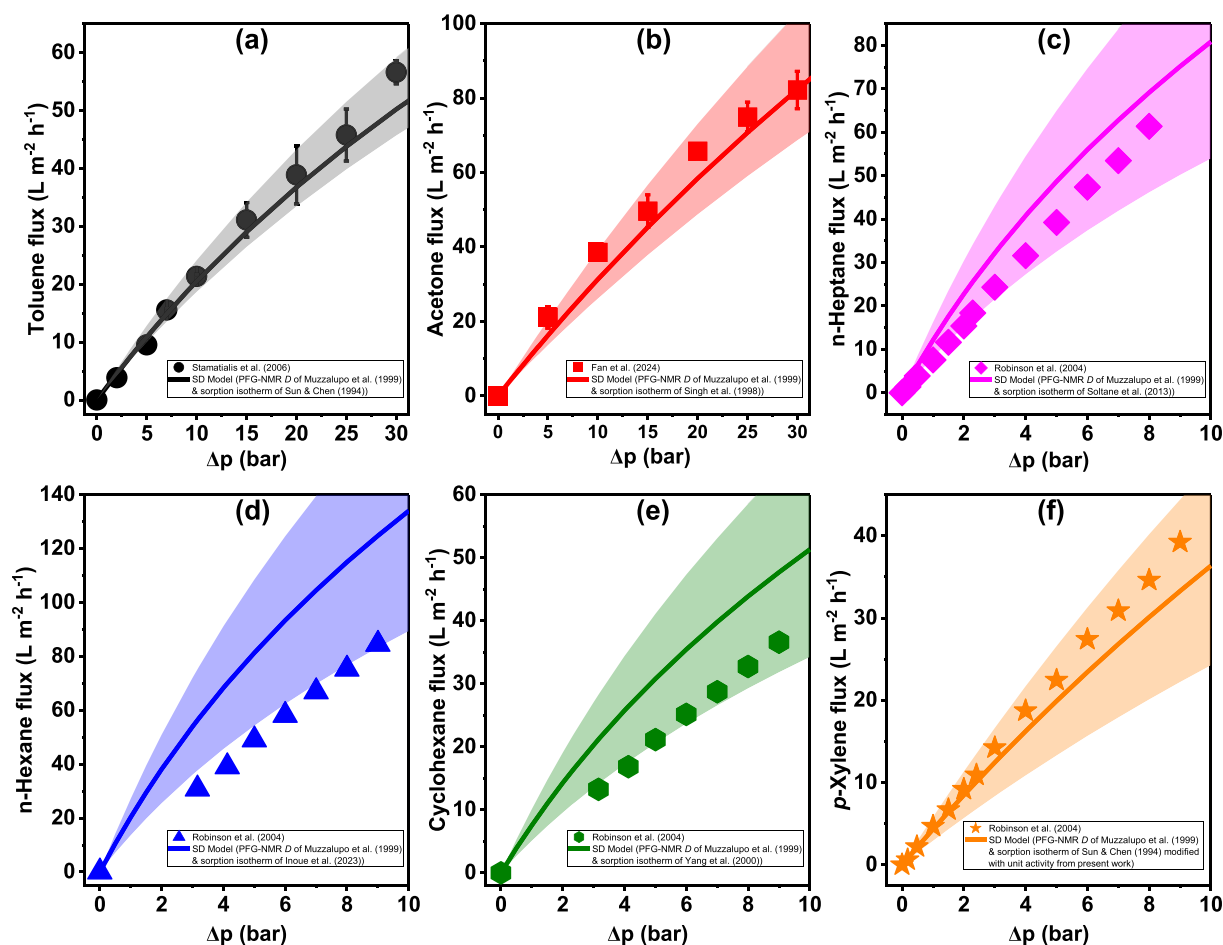


Fig. 3. Hydraulic permeation flux of various organic solvents in PDMS membrane as a function of transmembrane pressure using independent literature data for solvent sorption isotherm and PFG-NMR diffusivity estimation for SD parameterization for (a) toluene –sorption isotherm data of Sun & Chen, 1994 [41] and experimental flux data are obtained from Stamatialis et al., 2006 [58], (b) acetone –sorption isotherm from Singh et al., 1998 [42] and pure component permeation data of Fan et al., 2024 [15], (c) n-heptane –unit activity sorption uptake from Soltane et al., 2013 [30] and experimental flux measurement of Robinson et al., 2004 [59], (d) n-hexane –sorption isotherm of Inoue et al., 2023 [54] and pure component permeation data of Robinson et al., 2004 [59], (e) cyclohexane –unit activity sorption uptake from Yang et al., 2000 [55], and pure component data of Robinson et al., 2004 [59], and (f) *p*-xylene –sorption isotherm data of Sun & Chen, 1994 [41] modified by unit activity sorption uptake from this present study, and pure component permeation data of Robinson et al., 2004 [59]. The sorption data were fitted to the Flory-Rehner isotherm, and the diffusion coefficients of the aromatic, aliphatic, cyclic aliphatic, and ketone compounds in PDMS films were obtained from the PFG-NMR measurement of Muzzalupo et al., 1999 [36]. The shaded regions are based on the variability in the selective PDMS layer as obtained from the permeation studies, and where missing, a 20 % variation in film thickness is considered.

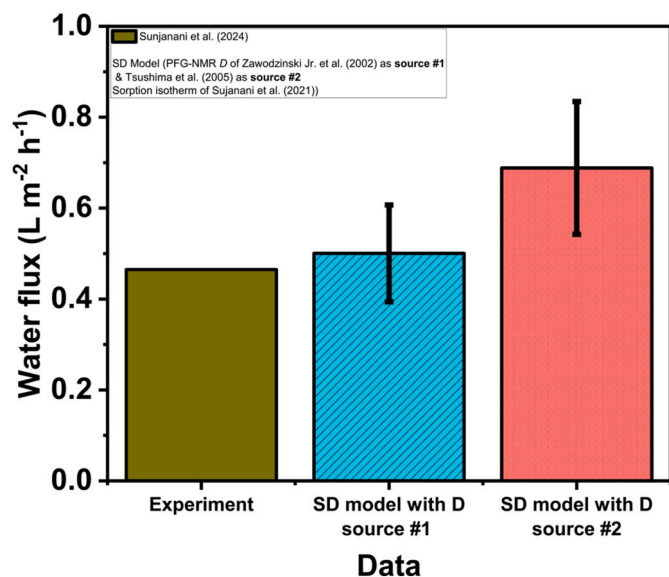


Fig. 4. Extension of the solution-diffusion (SD) model predictions to water transport through Nafion-117 membranes. The experimental data is that of Sunjanani et al., 2024 with water flux through a 200 ± 40 μm thickness of the proton exchange membrane using custom-made hydraulic permeation setup at a transmembrane pressure of 34.5 bar [60]. The “one-point” (unit-activity) sorption fitting was done using uptake data from Sunjanani et al., 2021 [53]. Two PFG-NMR measurement sources were considered for the water self-diffusivity in Nafion membranes (Zawodzinski Jr. et al. [56] and Tsushima et al. [57]) and are presented in Table S2 and section S3 of the supplementary information. The error bars in simulation results account for the influence of the membrane thickness variation on flux predictions.

(SD) framework using self-diffusivities from pulsed-field gradient NMR reported by two independent sources [56,57]. Despite the different data origins, both diffusivity inputs closely reproduce the measured water permeation rates through ionomer membranes, with deviations generally within experimental uncertainty. This agreement indicates that the SD model is robust to modest variability in the diffusivity estimate and that PFG-NMR-derived values are transferable to membrane transport predictions without additional fitting. This underscores the internal consistency and predictive power of the SD model, reinforcing its role as a unifying framework for mass transport in dense membranes.

Vapor permeation and pervaporation (PV) are other membrane transport processes that can be well-represented by the solution-diffusion model. The vapor permeation and pervaporation experiments of toluene through dense membranes of thicknesses (20–180 μm) were modeled via the SD equations given in Eqn. (16) and (17). The earlier sorption isotherm fitting of experimental data of Sun and Chen, 1994 was used for the prediction of toluene concentration within the membrane phase [41].

Fig. 5 presents the SD prediction of toluene fluxes across polymer membranes in three different membrane modalities, including OSRO, vapor permeation, and pervaporation. The SD model provides comparable flux prediction to experimental data across different upstream toluene relative saturations under the vapor permeation experimental conditions of Lue et al., 2008 [61]. In a two-phase membrane process involving upstream organic liquid and downstream vapor (i.e. near zero downstream activity), the versatility of the SD model was exemplified by comparably predicting the toluene pervaporation data of Heymes et al., 2006, [62]. This demonstrates that the solution-diffusion model can be utilized to predict a wide variety of diffusion-based transport modalities, provided the underlying parameters are known.

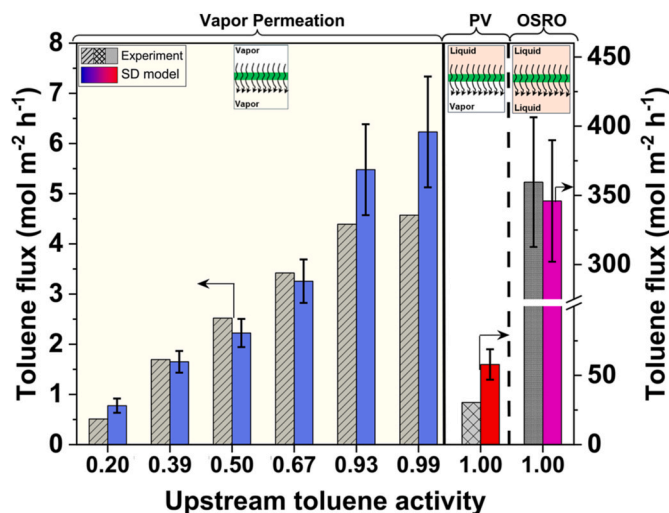


Fig. 5. The solution-diffusion mechanism provides comparable predictions to experimental fluxes under three different membrane processes involving liquid and vapor penetrants through dense (nonporous) selective membrane layers across three different experimental studies. The patterned gray bars represent experimental toluene fluxes, and the un-patterned blue, red, and pink bars denote SD predictions. The first (left) part of the figure shows the prediction of toluene vapor permeation through a 180 μm dense PDMS film compared to experimental data of Lue et al., 2008 [61]. The pervaporation (PV) data (middle part) represents that of Heymes et al., 2006, of toluene flux through a 20 μm PDMS layer thin-film composite membrane [62], while the OSRO data (right part) is taken at 20 bar transmembrane pressure from liquid toluene permeation of Stamatialis et al., 2006 [58]. SD parameterization was carried out using the toluene sorption isotherm of Sun & Chen, 1994 [41] and liquid-phase PFG-NMR diffusion coefficient of Muzzalupo et al., 1999 [36] by fitting it to the Mackie-Meares' model to obtain the concentration-sensitive diffusivities in vapor permeation and pervaporation processes [50]. Error bars for SD prediction indicate 20 % variability in the active membrane thicknesses. Experimental error bars were included if the original data source denoted the errors. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

5. Robustness of SD validation to varying sorption isotherm and uncertainty in swelling estimates

The choice of sorption isotherm could substantially affect the fidelity of SD flux predictions. To assess this sensitivity, Fig. S4 compares SD predictions generated with multiple literature sources of sorption data, with details in Table S1. Across all organic solvents examined, predictions based on full sorption isotherms show errors that are comparable to experiment, whereas the largest deviations arise when a single “unit activity” uptake value is used to create a surrogate isotherm. For toluene and acetone, isotherms that span a range of activities yield strong agreement and substantial overlap with measured fluxes, consistent with the need to capture nonlinearity in polymer–solvent interactions at intermediate activities. In contrast, for n-hexane, the increased deviation observed when using the full isotherm of Inoue et al., 2023 [54] likely reflects material and methodological differences, specifically the use of loosely crosslinked PDMS that was ground for uptake measurements, which can alter free volume, network dilation, and thus the sorption uptake of a solvent. For *p*-xylene, coupling the “partially-full” isotherm of Sun and Chen 1994 available up to a ≈ 0.5 [41] with our in-house unit-activity uptake produces predictions that are again consistent with experiment. Overall, access to a full isotherm improves SD accuracy and should be considered essential for making precise SD predictions.

In Fig. 6, we quantify how uncertainties in polymer swelling used to correct the dry-film thickness via Paul's relationship (Eq. (19)) propagate into the solution-diffusion (SD) model validation. For each

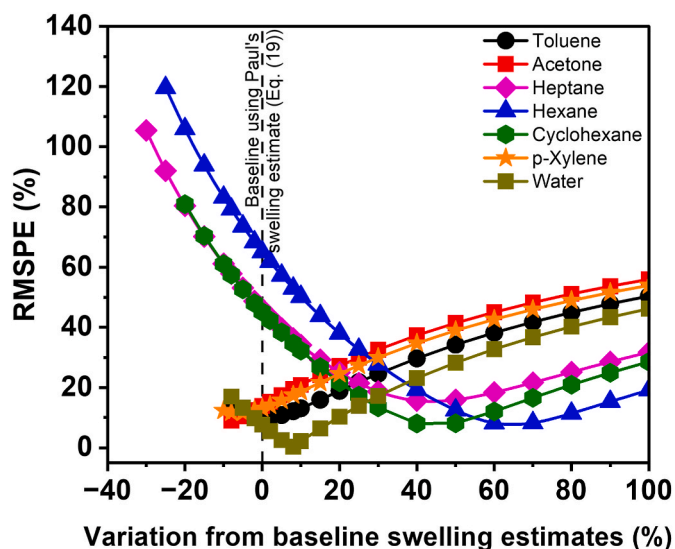


Fig. 6. The root mean square percentage error (RMSPE) of the solution-diffusion (SD) model predictions for the organic and water permeation in membranes with variation in the assumption of Paul's relationship for the estimation of selective layer swelling from the nominal thicknesses. The negative variations from our baseline swelling estimates are ultimately bounded by the dry thickness of the membrane.

penetrant, the root-mean-square percentage error (RMSPE) between predicted and measured fluxes was recalculated while systematically perturbing the baseline swelling estimate. The “U”-shape nature of the resulting error surfaces demonstrates strong sensitivity of SD predictions to the membrane swelling correction. Notably, the minima for the aromatic and aliphatic hydrocarbons occur at positive deviations (~20–50 %) from the baseline swelling, indicating that the baseline application of Paul's thickness correction [11], likely underestimates dilation for nonpolar penetrants (in other words, if the permeation data were fit to find the swollen membrane thickness, the thickness would be higher than that estimated by Paul's equation). In contrast, polar penetrants (e. g., water, acetone) exhibit broader, shallower minima closer to the baseline (0–20 %), implying comparatively lower sensitivity consistent with their smaller volumetric uptake and stronger sorption–diffusion coupling in the polymer. Across all cases, mis-specifying swelling by more than ± 25 % leads to rapid error growth, underscoring that accurate swelling quantification is also critical to SD prediction fidelity.

6. Conclusions

In this work, the solution-diffusion transport model is examined for its suitability for organic solvent and water permeation in nonporous polymeric membranes. This work seeks to shed more light on the assumptions and underlying mechanics of the SD model with a focus on testing its accuracy in predicting organic solvents and water transport using parameters from unrelated sources. Our analysis has shown that by combining two different equilibrium (sorption and diffusion) datasets

for a wide range of organic solvents (linear and cyclic aliphatics, aromatics, and ketones) and water in dense membrane films (and ionomer membranes) with permeation data through thin-film composite (or freestanding) membranes from independent labs, we were able to accurately estimate the membrane fluxes of these organic solvents and water during hydraulic permeation. Using a model that accounts for the concentration-dependence of the diffusion coefficient, we extend these predictions and comparisons to pervaporation and vapor permeation datasets, which also yield predictions with good accuracy. We also found out that Paul's swelling relationship is likely within reasonable estimates of actual membrane swollen thickness under permeation conditions. We underscore that the availability of a full isotherm is critical for accurate sorption coefficient estimation. This suite of comparisons and calculations – based on disparate experimental datasets for $\mathbb{P}$ (or flux), D , and $\mathbb{S}$ – underscores a key point of this manuscript: the fundamental mechanisms of transport in the solution-diffusion model can be independently measured, and broadly speaking, yield the result of $\mathbb{P} = D \cdot \mathbb{S}$. Of course, non-idealities such as concentration-dependent diffusion coefficients, non-linear isotherms, membrane dilation, and others must be properly taken into account to yield such a satisfying result, but the underlying mechanisms of solution-diffusion transport are well-understood and are – importantly – independently observable.

That said, there are many open questions remaining on the nature of molecular transport in membrane materials, especially in materials that blur the boundary between diffusive and viscous transport. Examples include solvent-dilated polymers of intrinsic microporosity, large pore carbon molecular sieves, multi-layered covalent organic framework structures, and other types of microporous materials that often exhibit multiple sorption and diffusion environments. Where possible, measurements of transport phenomena independent of permeation techniques should be pursued to deconvolute the various ways in which molecules and fluids move through polymeric and microporous systems.

CRedit authorship contribution statement

Hammed A. Balogun: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis. **Ryan P. Lively:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.memsci.2025.124731>.

Nomenclature

a_i	component's activity [dimensionless]
B	diffusional matrix [$s\ m^{-2}$]
c_i	molar concentration of component i [$mol\ m^{-3}$]

D_i	diffusivity of component i [$m^2 s^{-1}$]
D_i^{m0}	component i self-diffusivity in its organic or aqueous phase [$m^2 s^{-1}$]
$D_{i,self}$	self-diffusivity of component i [$m^2 s^{-1}$]
$\mathfrak{D}^{v,m}$	volume-based Maxwell-Stefan diffusivity [$m^2 s^{-1}$]
f	fugacity [Pa]
g	gradient pulse intensity [$T m^{-1}$]
J_i	molar flux of component i [$mol m^{-2} s^{-1}$]
K_i^G	gas-phase sorption coefficient of component i [$mol m^{-3} Pa^{-1}$]
N_i	flux of component i [$mol m^{-2} s^{-1}$]
N_1^m	penetrant molar flux [$mol m^{-2} s^{-1}$]
N_1^v	volumetric flux of component i [$m^3 m^{-2} s^{-1}$]
$\mathbb{P}_i$	permeability of component i [$mol m^{-1} s^{-1} Pa^{-1}$]
p	pressure [Pa]
p_i	partial pressure of component i [Pa]
p_i^{sat}	saturated vapor pressure of component i [Pa]
q	wavevector [m^{-1}]
R	gas constant [$J mol^{-1} K^{-1}$]
$\mathbb{S}_i$	sorption coefficient of component i [$mol m^{-3} Pa^{-1}$]
S	Pulse-field gradient-NMR signal intensity [<i>a.u.</i>]
T	temperature [K]
t	time [s]
$\hat{V}_i$	bulk liquid molar volume of component i [$m^3 mol^{-1}$]
$\bar{V}_i$	partial molar volume of component i [$m^3 mol^{-1}$]
x	molar fraction (dimensionless)
z	spatial dimension across membrane thickness [m]
δ	effective gradient pulse length [m]
ℓ	membrane thickness [m]
μ_i	component i 's chemical potential [$J mol^{-1}$]
γ	activity coefficient (dimensionless)
γ'	plasticizing coefficient [$m^3 mol^{-1}$]
γ^*	gyromagnetic ratio [$rad s^{-1} T^{-1}$]
Γ	sorption matrix (dimensionless)
Ψ	Pulse-field gradient-NMR signal attenuation (dimensionless)
ϕ	volume fraction in the polymer phase (dimensionless)

Subscripts

0	feed side of the membrane
f	feed
m	membrane or polymer
n	number of components
p	permeate

Data availability

Data will be made available on request.

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